

ST JOHN'S INTL, NL, Canada

WMO: 718010

Lat: 47.620N

Lon: 52.750W

Elev: 461

StdP: 14.45

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	10.3	-1.6	5.1	8.9	1.7	6.1	11.6	42.6	25.5	39.8	26.7	15.9	290	0.826

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	77.0	66.7	74.3	65.3	71.8	64.0	69.4	73.8	67.6	72.0	65.7	70.0	15.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.8	104.1	72.0	65.8	96.9	70.3	64.0	90.7	68.3	33.7	74.2	32.3	72.1	30.8	70.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.7	31.1	28.1	DB	3.4	82.4	3.9	2.2	0.6	84.0	-1.6	85.3	-3.8	86.5	-6.7	88.2	(4)
(5)			WB	2.3	72.5	3.8	1.5	-0.4	73.7	-2.7	74.6	-4.8	75.4	-7.6	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	25.6	24.5	28.0	35.5	43.1	51.4	61.5	61.9	55.0	46.3	38.5	30.3	(6)	
	DBStd	14.64	7.60	7.90	6.93	6.19	6.73	7.43	6.39	5.47	6.22	6.43	6.95	7.43	(7)	
	HDD50	4040	756	714	681	436	236	71	4	1	21	155	352	613	(8)	
	HDD65	8497	1221	1134	1146	884	678	410	142	125	304	581	795	1077	(9)	
	CDD50	1087	0	0	0	2	23	114	361	369	172	39	6	1	(10)	
	CDD65	69	0	0	0	0	0	3	33	28	5	0	0	0	(11)	
	CDH74	258	0	0	0	0	1	13	137	93	14	0	0	0	(12)	
	CDH80	14	0	0	0	0	0	0	8	5	1	0	0	0	(13)	
	WSAvg	13.8	16.3	16.0	15.2	14.0	12.6	12.2	12.2	11.5	12.6	13.4	14.5	15.7	(14)	
	PrecAvg	60.9	6.1	5.4	5.4	4.6	4.1	3.9	3.4	4.6	4.9	6.3	6.0	6.2	(15)	
(16) Precipitation	PrecMax	75.9	9.4	9.6	8.8	11.7	7.2	7.3	6.9	13.3	10.1	13.0	11.2	10.5	(16)	
	PrecMin	46.5	2.7	2.4	2.3	1.4	0.9	0.9	0.9	1.7	1.3	3.2	2.2	2.5	(17)	
	PrecStd	7.0	1.9	1.7	1.7	2.2	1.6	1.6	1.5	2.4	1.9	2.1	2.0	2.0	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	49.4	50.6	60.2	70.2	75.9	80.9	80.2	75.7	67.9	61.6	55.0	(19)	
		MCWB	51.2	48.0	47.5	53.7	59.4	64.8	68.6	69.0	67.2	64.4	59.1	53.6	(20)	
	2%	DB	45.7	43.5	45.2	54.1	64.6	72.1	77.5	76.6	71.1	63.1	57.3	49.8	(21)	
		MCWB	45.0	42.6	42.6	49.1	55.9	62.1	66.8	67.1	64.7	60.0	55.8	48.9	(22)	
	5%	DB	39.7	38.8	40.4	49.5	59.9	68.3	75.1	73.8	67.9	60.0	53.7	44.3	(23)	
		MCWB	38.9	37.7	38.2	45.4	52.8	60.1	65.9	65.1	62.8	57.3	52.2	43.7	(24)	
	10%	DB	35.5	34.8	36.8	45.2	55.2	64.5	72.3	71.2	65.0	56.4	49.7	40.1	(25)	
		MCWB	34.5	33.6	34.8	41.7	49.5	57.8	64.4	64.1	60.7	53.5	48.3	39.2	(26)	
	0.4%	WB	51.3	48.3	48.1	56.2	62.3	68.1	72.1	71.7	69.7	64.9	59.7	54.1	(27)	
		MCDB	52.3	49.4	50.3	58.8	66.9	73.2	76.8	77.0	72.6	66.8	60.8	55.0	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.0	42.7	43.3	50.6	57.7	65.0	69.7	69.2	66.4	61.1	56.0	48.9	(29)	
		MCDB	45.6	43.6	45.3	52.7	62.3	69.5	74.7	73.6	69.6	62.4	57.1	49.6	(30)	
	5%	WB	39.1	37.9	38.5	45.9	53.8	62.0	67.9	67.3	63.8	57.5	52.2	43.7	(31)	
		MCDB	39.6	38.6	39.7	48.4	58.3	65.9	72.7	71.6	67.0	59.3	53.6	44.3	(32)	
	10%	WB	34.7	33.7	35.0	42.0	50.0	58.6	65.9	65.4	61.3	53.8	48.5	39.4	(33)	
		MCDB	35.4	34.6	36.4	44.8	54.8	63.2	70.3	69.7	64.4	56.0	49.7	40.0	(34)	
	5% DB	MDBR	11.6	11.4	11.0	12.2	15.0	15.7	15.9	13.8	13.7	11.0	10.7	9.9	(35)	
		MCDBR	18.8	16.9	16.2	19.7	23.2	22.2	19.3	17.9	16.9	15.5	16.4	16.6	(36)	
		MCWBR	19.3	17.0	14.5	15.3	15.6	13.7	11.4	10.5	11.2	13.1	15.9	17.1	(37)	
		MCWBR	18.8	16.9	15.9	18.9	20.9	19.6	17.0	15.9	15.7	15.2	16.2	16.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.315	0.334	0.365	0.368	0.401	0.429	0.396	0.369	0.344	0.323	0.312	(40)	
		taud	2.359	2.433	2.413	2.370	2.424	2.356	2.317	2.421	2.469	2.493	2.463	2.373	(41)	
	Ebn at Noon		239	270	282	282	284	274	264	269	266	256	237	225	(42)	
		Edh at Noon	23	26	31	36	35	38	39	34	30	24	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		299	545	915	1249	1527	1615	1625	1372	1090	660	373	247	(44)	
	RadStd		23	41	73	125	124	126	120	88	69	61	50	21	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page